

Verified Papers and Gaps in IDS/IoT Research

Paper (Author, Venue, Year)	IF (Year)	Key reported metrics / findings (with citations)
Accelerating IoV Intrusion Detection (Çolhak <i>et al.</i> , CIIT 2025) [5†L59-L64]	– (conf.)	Using NVIDIA cuML vs CPU scikit-learn: up to 159× faster training and 95× faster inference on IoV IDS benchmarks [5†L59-L64] .
Improvement of Anomaly Detection (Benaddi <i>et al.</i> , IEEE GLOBECOM 2023) [38†L1291-L1297]	– (conf.)	Hybrid 1D CNN–LSTM on BoT-IoT: 99.20% accuracy , 0.80% false alarm (binary classification) [22†L127-L132] .
<i>ChatIDS: Explainable Cybersecurity</i> (Jüttner <i>et al.</i> , SECURWARE 2023) [41†L6-L9]	– (conf.)	Proposed ChatIDS using ChatGPT to explain IDS alerts. 4-page SECURWARE “best paper”; qualitative feasibility study showing potential security improvements (no numeric accuracy reported) [40†L55-L63] .
<i>Think Fast: IoT IDS with LLMs</i> (Jamshidi <i>et al.</i> , arXiv 2025) [45†L39-L43]	– (arXiv)	Edge-centric IDS + LLMs: up to 98% accuracy (six ML models), and LLM-based threat analysis with <1.5 s latency , <1.2 kB/prompt , <75 J energy use [45†L65-L69] .
BoT-IoT dataset (Koroniotis <i>et al.</i> , FGCS 2019) [79†L54-L58]	7.187 (2019) [83†L163-L170]	Large IoT botnet dataset paper. Authors: Koroniotis, Moustafa, Sitnikova, Turnbull. Published in <i>Future Generation Computer Systems</i> 100 (2019):779–796 [79†L54-L58] . Defined the Bot-IoT dataset (used as IDS benchmark).

- **Custom CUDA inference kernels (IDS):** *No specific publications found* on hand-written CUDA kernels for IDS model inference. Existing works use GPU libraries (e.g. cuML, TensorRT) rather than bespoke kernels.
- **LLM + IDS latency:** *No study isolates LLM overhead.* Aside from Jamshidi *et al.* [45†L65-L69] (above) which reports overall LLM response latency (<1.5 s), we found **no IDS+LLM work measuring extra latency costs** of adding an LLM.
- **Per-kernel GPU profiling:** *No IDS papers report Nsight/roofline analyses.* Literature on IDS does not include fine-grained GPU kernel profiling; this appears to be an unexplored gap.
- **TensorRT vs custom CUDA (<1M params):** *No direct comparison papers found.* We found no benchmarks comparing TensorRT to hand-written kernels for small (<1M) models in IDS or security contexts.

Sources: All above details are drawn from the referenced papers. For example, Çolhak *et al.* report the 159× speedup in CIIT 2025 [5†L59-L64] , Benaddi *et al.* report 99.20% on BoT-IoT [22†L127-L132] , Jüttner *et al.*

present ChatIDS at SECURWARE 2023 【40†L55-L63】 , Jamshidi *et al.* (2025) report the 98% accuracy and LLM metrics 【45†L65-L69】 , and Koroniotis *et al.* (FGCS 2019) is cited by UNSW 【79†L54-L58】 with IF from JCR 【83†L163-L170】 . (No gaps references found for the null cases, confirming those research gaps.)
